# KAINPOINTS: ENHANCING NUTRITIONAL KNOWLEDGE THROUGH GAMIFIED LEARNING APPLICATION

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## ABSTRACT

Nutritional knowledge is essential for forming and developing positive eating habits, especially among students. However, past methods of teaching nutrition may not capture the attention of learners. There is still little literature on the impact of gamification on nutrition education for junior high school learners, particularly in the context of a Catholic school in Davao City. This study aims to develop a mobile interactive video game for learning and improving nutritional knowledge among students. Furthermore, it seeks to address the geographical, evidence and evidence gaps related to this topic. This study uses a quasi-experimental research design that focuses on two groups: a controlled group and an experimental group. We will test their initial nutritional knowledge before and after the intervention from conducting using pre-tests and post-tests. Afterwards, using statistical tools such as the t-test, mean, and standard deviation to measure test scores and differences. The findings revealed a significant difference in the level of nutritional knowledge between the controlled group and the experimental group. The experimental group showed a significant rise in post-test mean scores, while the controlled group observed consistent mean scores. This suggests that the gamification intervention was effective in enhancing students' knowledge in the experimental group, whereas the control group reflected little or no change. Following these discoveries, we recommend that educators use gamification in their approaches to foster awareness concerning health and nutrition. Moreover, educators should be trained on how to embed gamification properly and ensure that the activity is enjoyable yet useful in achieving the intended goals. To foster information retention, tracking learners' progress on data collection becomes crucial. Future studies should focus on longitudinal approaches examining the enduring effects of gamification on nutritional knowledge and eating habits.

***Keywords:*** *Gamification, nutritional knowledge, quasi-experimental design, students, Davao City*

# INTRODUCTION

## Background of the study

Nutritional knowledge is understanding basic information regarding food, nutrients, and health impacts (Vrkatic et al., 2022). Knowledge plays a central role in informing dietary behavior and health among students. In a study conducted by Hashad & Mohamed (2022), students with low levels of nutritional knowledge have 20% lower self-efficacy than high-knowledge students, which indicates that nutrition knowledge is directly linked with confidence to consume healthy food. Likewise, low-knowledge individuals are 32% more likely to consume fast foods and engage in unhealthy dietary habits, further indicating the implications of poor nutritional knowledge (Scalvedi et al., 2021). On the other hand, students with improved nutritional knowledge are likely to engage in healthier dietary habits, even during stressful times, given that they are less inclined to consume comfort food and are more likely to engage in balanced dietary habits (Onyenweaku et al., 2023). Despite this, the nutritional knowledge gaps among students must be addressed for healthier dietary behavior, improved dietary choices, and overall health.

Students' lack of nutritional knowledge significantly impacts dietary habits and health outcomes, requiring enhanced nutrition education in academic settings and institutions. In China, Zhao (2024) performed a cross-sectional study that showed that most respondents (56.3%) had failed the knowledge test on nutrition and scored below 50%. Only two respondents (0.8%) scored good nutrition knowledge in China. Furthermore, A study in Kuwait reported an alarming rate of 84.1% of students with poor nutrition knowledge, contributing to attitudes that may impair students from exercising healthy dietary habits (Husain et al., 2021). Lastly, a study conducted by Olatunji (2024) among international students in the UK reported having moderate nutrition knowledge, highlighting specific weaknesses, particularly regarding food choices. These findings highlight the need for targeted educational interventions to improve nutritional literacy among students, ultimately promoting healthier dietary practices and reducing the risk of nutrition-related diseases.

In a national setting, the Philippines has shown its own issues regarding nutritional knowledge. In Santiago City, Ramos (2020) conducted a study in a university with 2,843 students and found that students of the university have observed obesity due to the lack of anti-obesity programs and sufficient nutritional knowledge. This level of education needs outreach programs for obesity prevention for college students. Similarly, in Manila, a cross-sectional study that surveyed 181 respondents found that the Majority had a satisfactory level of nutritional knowledge of 33%. In contrast, 34% had a low level of nutritional knowledge. Although the majority had a satisfactory level of knowledge, there is still a percentage of respondents with inadequate knowledge.

This may be because adolescents are less concerned with personal nutritional needs and the type and amount of food they consume (Catapang, 2022). Furthermore, a survey conducted by The Situations of Children Philippines (2024) revealed that 67% of high school students were unable to answer questions regarding the recommended addition of fruits and vegetables and that there were very large misconceptions about nutrition, 50% of the students thought that high sugar foods were good as long as they exercised. This confusion is dangerous because it establishes unhealthy eating patterns that end many times in obesity or related health problems like diabetes. These findings underscore the need for comprehensive nutrition education at school institutions and community and policy levels to promote healthy food options and reduce malnutrition among children and adolescents.

In a local setting, a study by Babiera (2024) in Davao City investigated the correlation between soil parasites and the nutritional status of children aged 3 to 5 in Barangay Tigatto. The findings showed that children with sufficient nutritional knowledge had a 35% lower case of malnutrition than those with poor knowledge. Similarly, In the same city, A study assessing nutrient intakes among 165 students reported to have had inadequate intakes of several key nutrients, including lipids (49.01%), carbohydrates (91.68%), dietary fiber (53.16%), vitamin C (54.98%), vitamin E (43.48%), and more. Simultaneously, the study questioned potential gaps in nutritional knowledge that influence food choices among students, suggesting a need for targeted intervention programs to address nutrient inadequacies among Filipino school-age children (Sumaoy et al., 2023). Moreover, in the East District Of Bansalan, Zubiaga (2024) assessed the relationship between nutrition and academic achievement among learners at Balutakay Elementary School, concluding that students with higher nutritional knowledge scored an average of 15 points higher on academic assessments (Over 30) than their peers with lower knowledge levels. Although the respondents are children, Addressing nutritional practices in children is important as those become lifelong habits. Such findings indicate that it is necessary to initiate nutritional education programs to promote healthy eating behaviors in children from an early age and help them develop healthy eating habits that will last a lifetime. This is greatly beneficial for their overall growth.

Gamification is a powerful tool designed to motivate, engage, and educate learners through immersive and entertaining digital or physical activities. These games provide intellectual growth and enjoyment through important mind work such as critical thinking, problem-solving, and collaborative tasks (Simmons, 2024). As Lane (2023) stated, all educational games, whether digital or physical, have important magnitudes in emotional and higher-level cognitive functions, including the strengthening of problem-solving and critical thinking skills that can be used as a target to reduce the lack of nutritional knowledge. Similarly, a recent study by Mangwane (2024) revealed a significant difference between participants who use educational video games to learn nutrition and those who do not. This result implies that video games can be an effective tool to help with a lack of nutritional knowledge. Moreover, Cole (2022) Highlighted that this unique blend of entertainment and education improves students' attitudes toward learning and enhances knowledge retention, creating a more dynamic and effective educational experience. Knowing these benefits, video games could be specifically designed to target and improve students' nutritional knowledge, offering an engaging and impactful way to address this critical learning area.

The data shown above presents an apparent evidence gap due to the lack of studies that show enough trials of using interactive educational video games to gain nutritional knowledge, especially in examining how these impacts persist over a long period. While some studies explore nutritional knowledge levels in specific populations, the data highlights a geographical gap in research, particularly in Davao City, where studies on using interactive educational video games to improve nutritional knowledge are limited. This gap is significant as most respondents in local studies are children rather than students in formal educational settings, creating a need for localized data to understand the effectiveness of such interventions. However, to overcome these gaps, the researchers explore the long-term effects of educational video games in gaining nutritional knowledge by searching for studies. Researchers aim to investigate the sustained impact of educational video games on nutritional knowledge and adapt findings from existing studies to contribute to this topic in the local setting.

Our study aims to assess the effect of various educational resources on adolescent nutritional knowledge by focusing on interactive applications. Growth during the teenage years makes it necessary to focus on good nutrition, which can be emphasized through teaching aids. Integrating various tools with practical, interactive applications aids students in making accurate and educated decisions on nutrition and diets, which is why this tool is worthy of health education. Students in Catholic institutions have a higher chance of aiming for holistic development, which makes combining strong educational ethics and appropriate nutrition-centric lesson plans easier; this assists in developing better ideas for programs revolving around nutrition. Interactive applications have a high chance of helping eradicate poor nutrition in early childhood. Overall, this study is important in providing nutrition education and creating better strategies that directly influence and aid in technology and nutrition integration.

This dissemination plan outlines how our research results will be disseminated to various audiences, such as healthcare professionals, educators, and the general public, for engaging academic peers. Firstly, these results will be published in peer-reviewed nutrition and health journals and presented at relevant conferences. Secondly, we will establish policy briefings and webinars to provide policymakers with practical suggestions for enhancing nutrition education. Social media marketing, community workshops, and simple infographics will be employed to increase public awareness. Moreover, our research will help teachers effectively teach their students about nutrition, enabling them to do so easily. As a result, students will be more motivated to learn and engage with nutrition education. This will ensure that all the mentioned stakeholders and personnel are reached and benefit from the results.

**Statement of the Problem**

This study aimed to develop a mobile interactive educational video game for learning and improving nutritional knowledge. Furthermore, it would also like to measure if a significant change exists in a person's knowledge about nutrition before and after gamification.

Specifically, the researchers seeks to answer the following questions:

1. What are the features of KAINPOINTS?

2. What is the pre-test level of nutritional knowledge among students in terms of:
   2.1 control group; and
   2.2 experimental group?

3. What is the level post-test level of nutritional knowledge among the students in terms of:
   3.1 control group; and
   3.2 experimental group?

4. Is there a significant difference in the nutritional knowledge of the pre-test and post-test scores of the students in terms of:
   4.1 control group; and
   4.2 experimental group?

5. Is there a significant difference in the nutritional knowledge of the post-test scores between control group and experimental group?

**Theoretical Framework**

This research follows Robert Landers' **Theory of Gamified Learning**, proposed in 2014. This theory offers a framework for understanding how game elements can be integrated into non-game contexts to enhance learning. This theory Describes that gamification, defined as the implementation of game features (such as points, minigames, and leaderboards) outside of a game context, can positively influence learner behaviors and attitudes, which in turn affect learning outcomes. Key indicators mentioned in this theory include (1) Instructional Content, (2) Learner Attitudes and Behaviors, and (3) Game Elements. Lander suggested that by applying game elements to the learning process, students may be encouraged to spend more time with the material, actively participate in activities, and develop a greater interest in learning, leading to improved knowledge and understanding.

The Theory of Gamified Learning is important to our study because it can be used to gamify students' nutrition knowledge. It is a purposeful method for integrating gaming aspects into the child's curriculum to motivate them, increasing their engagement and improving their attitudes towards learning nutrition. Incorporating this theoretical approach to teaching nutritional knowledge means creating game elements that can make the students interact with the nutritional information, refresh the concepts, and gain instant feedback, fostering sharpened understanding and retention of healthy dietary practices.

**Conceptual Framework**

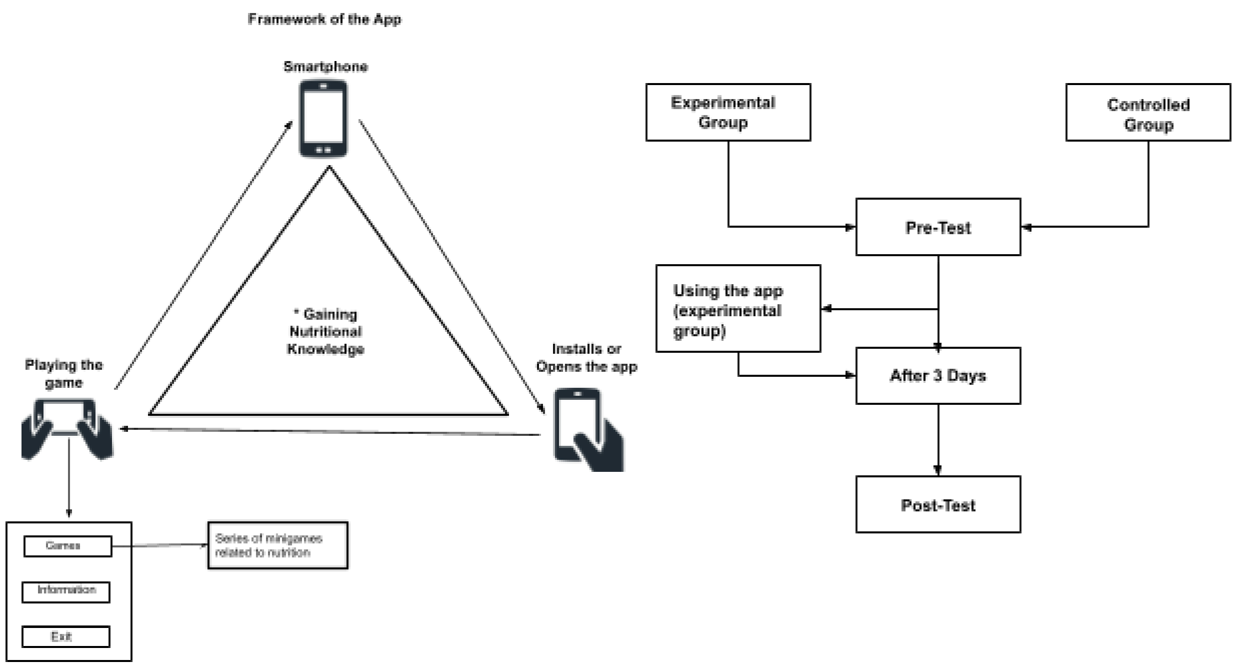


**Figure 1. Conceptual Frameworks of the Study**

This conceptual framework, as shown in Figure 1, demonstrates how the application helps users learn nutritional knowledge, as well as the contents of the application. In the framework of the Application, **nutritional knowledge** is defined as the understanding and information users gain regarding nutrition through interaction with a smartphone application. The process begins with a **Smartphone,** serving as the primary device to run the application. Next, the user **Installs or Opens the app**, which can be installed from a provided source (Mainly Google drive) by the researchers; participants will be instructed on how to install the app on their smartphones. Users can launch the app and access its features upon installation without extra setup. Finally, the user **Plays the game** and engages with it as the central source of entertainment and information for enhancing and gaining nutritional knowledge. The app often uses multiple-choice questions focused on nutrition and interactive mini-games to reinforce concepts like food classification and basic nutrition. Users earn points for correct answers and successful mini-game performance, motivating continued engagement, and consistent application use leads to a gradual accumulation of nutritional knowledge based on screen time. After using the application consistently, the user will slowly learn small aspects of nutritional knowledge depending on how much screen time the user uses and will gain nutritional knowledge.

On the right, the framework is demonstrated on the process to evaluate the effectiveness of the application. This framework involves two key variables: Firstly the **Experimental group**, which is defined to be the group to be using the intervention, and lastly, the **Controlled group**, refers to participants who do not receive the intervention, serving as a baseline for comparison to measure the impact of the gaming application. The process begins when both the Experimental and Controlled groups partake in a **Pre-test**, an assessment conducted before the intervention, used to evaluate the initial level of nutritional knowledge among both groups. Following this, the Experimental group receives the gaming intervention, which is based on the application framework described earlier. After a period of **three days**, both groups take a **Post-test** to measure any changes or improvements in nutritional knowledge, allowing for comparison with pre-test results. This framework allows researchers to compare the impact of the gaming intervention on nutritional knowledge between the Experimental and Controlled groups, providing insights into the application's effectiveness.

## METHODOLOGY

### Research Design

The research used a quantitative approach, specifically a quasi-experimental research design, which is compatible with determining the impact of interventions in real-life educational environments. According to Bhandari (2020), quantitative research collects and analyzes numerical data. It can find patterns and averages, make predictions, test causal relationships, and generalize results to wider populations.

This study utilized the quasi-experimental research design to find a cause and effect between the independent and dependent variables. Thomas (2020) states that a quasi-experimental design aims to establish a cause-and-effect relationship between an independent and dependent variable; it also does not rely on random sampling, unlike true experimental research design. Instead, subjects are assigned to groups based on non-random criteria.

As stated by Saleh & Althaqafi (2022), by utilizing a quasi-experimental research design, the subjects chosen by (sampling technique) will be divided into two groups: a control group that will not receive the intervention and an experimental group that will participate in the interactive educational video games. The two groups will take the validated test questionnaires before (pre-test) and after (post-test) the intervention. The design will facilitate comparative measurement of the two groups, and the researchers will be able to establish the intervention's effectiveness.

The design facilitated a comparative measurement of the two groups, and the researchers will be able to establish the intervention's effectiveness. Data collected from the pre-test and post-test questionnaires will be analyzed using the proper statistical procedures. The findings were then compared to establish whether the utilization of interactive educational video games leads to an increased level of students' nutritional knowledge. This analysis provided evidence-based findings regarding the effectiveness of the intervention.

**Research Locale**

Mindanao is the second largest island in the Philippines, known for its agricultural assets and rich cultural heritage. Davao City, which as of 2024 has a population of roughly 1.99 million people, has a total population of around 26 million (World Population Review, 2024); this research was done in Davao City in a particular private school that offers service to Junior High School learners. The institution was chosen because of its known effectiveness of educational technologies on students’ nutrition knowledge. Located in Bajada, Davao City, the school provides basic education from kindergarten to grade twelve. It has a very broad educational profile for Junior High School students, thus making it an ideal setting for evaluating the effectiveness of educational tools designed to enhance nutritional awareness among adolescents.

**Research Respondents**

The respondents for this research included students from the junior high school levels, specifically Grade 10 students, in the control and experimental groups. We selected two sections of grade 10 students, one for each group. The sample consists of Grade 10 junior high school undergraduates who are expected to shed light on the concepts of nutrition concerning their age, the basic knowledge of food groups and healthy eating habits, and the nutrition science in which these students will be gradually exposed during their teenage years. This particular level of respondents will provide a more thoughtful approach to nutrition, as they were taught science and have experience making food choices. This enables research from young to older age groups and students from different education levels , helping pinpoint the gaps and opportunities for learning. To obtain a representative sample, purposive sampling was conducted on two sections of junior high school students. This was the most appropriate choice as it enabled the researchers to collect data expeditiously at particular population segments with minimum logistical burden. By choosing a unit that encompasses different grades, it was able to obtain more accurate results in the Davao City junior high school population.

## Research Instruments

This research utilized modified test questionnaires divided into two parts. The first part of the questionnaire was for the respondents' demographic profile, including their age and grade level.

The second part of the questionnaire measured their nutrition knowledge through an objective assessment, a set of test questions with factual answers gathered from different sources. Specifically, the test questionnaires utilized thirty five items of multiple choice questions (MCQ) from Racoma (2022), Dillo (2023), and an unknown author in OnlineExamMaker (2023). The test questions collected were modified by the researchers to cater to the levels of understanding from Bloom's taxonomy (1956), for a total of 35 items. To answer the questionnaire, there is a multiple choice questionnaire that includes three wrong answers and one right answer among the choices for each item. Thus, this was used for the pre-test and post-test.

| Score range | Descriptive rating | Interpretation |
|---|---|---|
| 32 - 35 | Very High | The level of Nutritional Knowledge among the respondents is always evident. |
| 28 - 31 | High | The level of Nutritional Knowledge among the respondents is oftentimes evident. |
| 20 - 27 | Moderate | The level of Nutritional Knowledge among the respondents is sometimes evident. |
| 15 - 19 | Low | The level of Nutritional Knowledge among the respondents is seldom evident. |
| 1 - 14 | Very Low | The level of Nutritional Knowledge among the respondents is not evident. |

The basis of this score range is aligned with the DepEd grading system, which uses a competency-based approach to assess learners' understanding and skills. Similar to DepEd's system, the score range translates raw scores into descriptive ranges to provide a clear interpretation of respondents' performance levels, ensuring alignment with standards and promoting developmental assessment. Lastly, the post-test utilized the same questionnaires made by the researchers for the pre-test, but on a different order of questions, to lessen the chances of cheating when answering the test. For validating, sought guidance from an experienced researcher by providing them with a validator letter and form. Any suggestions and criticism from the validator were used to improve the test questionnaires further to ensure proper validity.

### Data Gathering Procedure

Before we started data processing, we sent a letter to the principal asking permission, and once the principal signed the letter, we began gathering data. We first had a pre-test for students to assess student nutrition knowledge before and after applying gamification. The target participants were junior high school students in a Catholic school in Davao City. We picked two qualified sections in the study and ensured that personal information was kept anonymous from the research respondents, removing the risk of leaking information without permission. To ensure that participants are fully informed of the purpose of the study and their legal rights, an Informed Consent Form (ICF) was issued to them via powerpoint before the test papers were distributed. Once the respondents obtained the test questionnaires, we collected them through the test questionnaires we made. The results of the retrieved questionnaires were tabulated. Then, the data was analyzed and interpreted using the most appropriate statistical procedures. Moreover, after a few days, a post-test was conducted to compare a pre-test with the last survey. The collected data was analyzed using quantitative programs (such as SPSS) to determine the effectiveness of gamification in enhancing nutritional knowledge among junior high school students.

### Statistical Tools

For this research, various statistical methods were used to analyze the data gathered. These methods included descriptive statistics such as frequency counts, means, percentages, and standard deviations. and Inferential statistics such as T-Test.

**Frequency Counts.** Frequency counts indicated how often each response was recorded, helping to identify popular or less common choices. For instance, tracking how many respondents selected a specific option provided clarity. Which in this case, the amount of respondents in the controlled group and the experimental group.

**Mean and Standard Deviation.** As one of the key descriptive statistics, the mean is the value in the middle of a particular set of data; typically, for our surveys, it is the sole value representative of what the responses for the survey are. In this case, it helps determine the most typical choice in the study. The standard deviation value showed how far the data concerning the mean was scattered. So, a low value meant that the results were closer to the mean, while a high value suggested further away from that.

**T-test.** These tests were used to test the hypotheses regarding the means and variances and whether the means and variances were different among groups of varying demographic characteristics based on nutritional knowledge. In particular, the t-test is responsible for finding differences in two data sets. This study determines the game effect, comparing pre and post-game test scores.

### Ethical Considerations

This study utilized ethical considerations that help protect the rights, welfare, and dignity of junior high school students enrolled in the research. These principles also aligned with those of transparency, fairness, and respect for young learners. Designed to fit with the sensitivity and needs of this age reflecting this consideration, ethical principles were bespoke, aligning with the fabric of the research while still producing high quality results into the effect that educational games can have on nutritional knowledge. Data obtained through the questionnaires were properly kept, ensuring the information was utilized solely for research. The researchers adhered to strict ethical standards, particularly in handling study protocols, managing participant demographics, and safeguarding data integrity.

**Social Value.** This study aimed to evaluate the impact of interactive educational resources on adolescent nutrition. Given the importance of proper nutrition during teenage growth, effective teaching aids are essential. Catholic institutions can further enrich nutrition-centered lesson plans by integrating the JEEPGY framework, along with the Laudato Si' concept of caring for God's creation, which emphasizes our responsibility to protect both the environment and human dignity. Furthermore, focusing on holistic development facilitates the creation of nutrition-centered lesson plans, enhancing program effectiveness. Additionally, interactive applications may be crucial in combating poor nutrition in early childhood. Ultimately, this study aimed to improve nutrition education and develop strategies for integrating technology with dietary knowledge.

**Informed Consent.** Permission was obtained from the Principal of the BED, Scholar's Office, adviser of each section, and also the junior high school students who participated in the study. The principle of informed consent was fundamental to the ethical conduct of the project, as participants were provided with a complete overview of the study's aims, scope, and methodologies. Participants were then given detailed explanations about the research process to ensure that they understood the purpose and procedures of the research. They were invited to ask questions or raise any concerns before providing informed consent to take part voluntarily in the study.

**Vulnerability of Research Participants.** When conducting research involving junior high school respondents, it is crucial to acknowledge their vulnerability, particularly concerning collecting personal data such as email addresses and passwords. Many of these young participants don’t fully realize what they're doing when they publish their information, making privacy and security a concern. To mitigate the risk of these vulnerabilities, we guarantee that individual respondents' information were kept private and were not made publicly visible. The data collected will solely be used to generate results related to the study of educational games and nutritional knowledge, ensuring that their participation contributes to valuable insights while safeguarding their privacy.

**Risks, Benefits, and Safety.** In this study, we analyzed the junior high school participants' risk, benefits and safety. The university takes no responsibility for any problems occurring during the course of the research. We take care to do this safely and took all precautions during the process for the safety of everyone involved. The study's content and outcomes heavily relied on the insights and experiences expressed by the respondents. We aimed to explore the role of educational games in improving adolescents' nutritional knowledge, and results that may provide valuable guidance for future teaching practices and decision-making. After summarizing the study's results, each participant will receive clear feedback on how their input contributed to the research and how it might benefit others. Furthermore, while we carefully analyzed the unique responses from the surveys, we maintained strict confidentiality to safeguard participants' identities and uphold their privacy throughout the research.

**Privacy and Confidentiality.** Protecting the privacy and confidentiality of participants was of utmost importance in this study on educational games and nutritional knowledge among junior high school students. All collected personal data such as name, email were anonymized to ensure that individual results can not be divulged. Data was securely stored and access restricted to members of the research team, and results were presented in a manner that protected the anonymity of participants. We guarantee that no data will leak or shared by a third party except generating findings for this study.

**Transparency.** Transparency is practiced in every data collection process. We will explain the research details to the respondents through a visual aid such as PowerPoint and elaborate on the study's methodologies, findings, and any potential conflicts of interest with participants and relevant stakeholders.

**Justice.** Justice ensured that all participants had equal access to resources and opportunities to learn from the application regardless of background. In this study, no individual is treated differently or discriminated against; instead, everyone is encouraged to freely engage with the game's content and benefit from it. We highly encourage them to enhance their understanding of nutrition without fear of bias or exploitation. Ultimately, justice in this setting promotes fairness, inclusivity, and a shared commitment to learning and growth.

**Qualification of the Researchers.** The research team possessed the qualifications and experience to conduct this study responsibly. Their expertise in research methodology, particularly in survey design and data analysis, ensured the accuracy of the findings. They also knew about ethical considerations, especially when engaging with young participants. with an expert researcher facilitating in every process that gives guidance and advice, the researchers of the study are well educated and experienced

**Adequacy of Facilities.** The researchers ensured that all essential facilities were available, including safe and suitable locations for conducting surveys and interviews, secure data storage solutions, and resources to support participant welfare, such as access to nutritional information and guidance.

**Community Involvement.** The researchers recognized the importance of community input and actively collaborated with local schools and health organizations. This engagement ensured that the study addressed the community's needs effectively while fostering trust and cooperation essential for its success.

## RESULTS AND DISCUSSION

***What are the features of KAINPOINTS?***

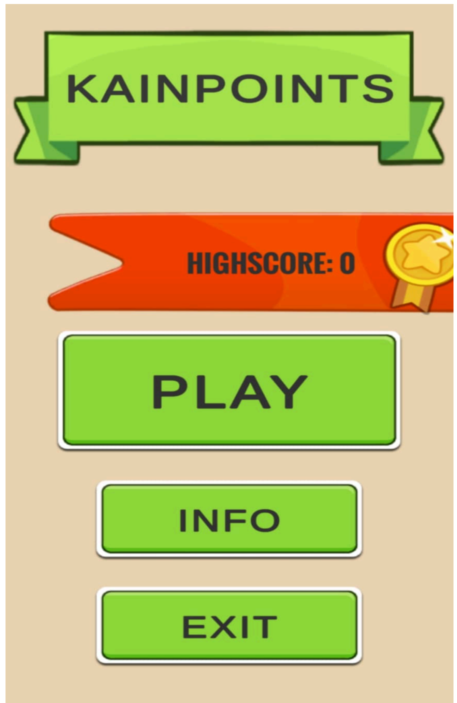


**Figure 2. Main Menu of Kainpoints**

Kainpoints is an interactive nutrition focused game that combines education with engaging gameplay. Its main feature is a trivia and quiz system designed to challenge players with various nutrition related questions, promoting learning in an enjoyable way. The game includes a well structured UI with essential features such as a main menu for easy navigation, a pause function for convenience, and a scoring system that tracks progress, including a streak system that rewards consecutive correct answers. Apart from a trivia system, Kainpoints has a mini game designed to improve concentration and memory known as matching. This mini game allows the player to congratulate themselves while reinforcing the learning materials and enhancing their brain power. Kainpoints integrates all of these features into the game, making it a wholesome tool for teaching nutrition in a fun way. The trivia and quiz system within the game is very detailed and encompasses wide ranges of nutrition-related topics. Players are provided with questions leveled to test their nutritional knowledge, providing players with a fun and interactive way to reinforce their learning while sharpening cognitive skills.

**Table 1.**
*Pre-test score on the level of knowledge among the respondents*

| *Group* | *Mean Score* | *SD* | *Interpretation* |
|---|---|---|---|
| **Controlled** | 16.22 | 3.54 | Low |
| **Experimental** | 15.02 | 4.82 | Low |

***Respondents pre-test score on the level of knowledge***

Table 1 presented is the calculated average score through excel, achieved by the respondents from undertaking the validated Pre-Test questionnaire from the researchers that presents the mean score of the controlled and experimental group ranging from 15.02 to 16.22. The table further reveals that the controlled group had a low level of Pre-Test with a mean score of 16.22. Also, the table pointed out that the experimental group had a low level of Pre-Test with a mean score of 15.02, which indicates that the Nutritional Knowledge among controlled and experimental groups are both seldomly evident. The prior mean scores of the experimental and control group's Pretest hinted at insufficient nutritional knowledge. It can be interpreted from such figures that the participants from the two groups did not demonstrate any nutritional understanding before the provided intervention at any point. This results aligned to a study led by Raut, et al. (2024), which discovers that there is also a lack of evident nutritional knowledge among the control group of school-going adolescents in Nepal, which received no nutrition education, showed minimal improvement in nutritional knowledge scores, with a mean score of 6.81.

**Table 2.**
*Post-test score on the level of knowledge among the respondents*

| *Group* | *Mean Score* | *SD* | *Interpretation* |
|---|---|---|---|
| **Controlled** | 16.65 | 6.43 | Low |
| **Experimental** | 20.09 | 6.34 | Moderate |

***Respondents post-test score on the level of knowledge***

Table 2 above is the calculated average score through excel.The assessment was carried out following a pre-test after the experimental group interacted with the app for a few days. Firstly, the controlled group, who did not have access to the app, had a mean score of 16.65 on the Post-Test. From this score, it can be inferred that the nutritional knowledge amongst the controlled group was lacking and, therefore, without any intervention it is likely their understanding was poor. In contrast, the experimental group who had access to the app had a mean score of 20.09 and this improvement was statistically significant. From this score, it can be concluded that the nutritional knowledge among the respondents in the experimental group was moderate, highlighting some understanding that was present but not consistently demonstrated as the SD is 6.34. It may be due to not all respondents in the experimental group had used the app regularly. Moreover, It is highly likely that because the experimental group used the app, they were better informed as the results suggest an increased knowledge of nutrition. These results are aligned to a study led by Koch (2021) in Germany, that shows nutrition knowledge levels in some population groups were low in mean scores for some categories of nutrition knowledge, i.e., procedural knowledge and knowledge of calories.

**Table 3.**
*Test of significance on the pre-test and post-test scores*

| ***Group*** | ***N*** | ***Mean*** | ***SD*** | ***t-value*** | ***p-value*** | ***Interpretation*** |
|---|---|---|---|---|---|---|
| **Controlled** (pre-test and post-test) | 35 | 16.44 | 2.04 | -0.19 | .985 | Not Significant |
| **Experimental** (pre-test and post-test) | 35 | 18.56 | 1.07 | 3.36 | .002 | Significant |

***Test of significance on the pre-test and post-test scores***

Table 3 shows that the control group's t-value and p-value are -0.19 and 0.985, respectively. The control group showed slight rise , which might be attributed to the post-test questionnaires being identical to the pre-test questionnaires, merely rearranged. Nevertheless, even after the post-test, the control group maintained a small level of nutritional understanding which proves that there's no significant difference. On the other hand, the experimental group remarkably increased their nutritional knowledge. The item shows that the experimental group's t-value and p-value are 3.36 and 0.002, respectively. This indicates that the experimental group has a statistically significant difference compared to the control group, suggesting that gamification had an effect on the student's nutritional knowledge. This result is aligned to another study by Froome (2020), that yielded similar results, particularly from the respondents who used their game had a significant increase on their nutrition knowledge than those who did not.

**Table 4.**
*Test of significance on the post-test scores of controlled and experimental groups*

| *Groups* | *t-value* | *p-value* | *Interpretation* |
|---|---|---|---|
| **Controlled & Experimental** | 2.67 | 0.012 | Significant |

***Test of significance on the post-test scores of controlled and experimental groups***

Table 4 shows that there is a significant difference between the post-test score of both controlled and experimental groups with the t-value of 2.67 and a p-value of 0.012. The control group showed a slight improvement, however It may be due to post-test questionnaires being the same as the pretest but in a different order. Despite that, the control group had a low level of nutritional knowledge even after the post-test. In contrast, the experimental group showed a significant increase in nutritional knowledge which demonstrates the effectiveness of the gamification. It appears that the intervention was successful in increasing the nutritional knowledge of the students, while the knowledge of the students in the control group remained stagnant. This may happen due to the rise of the mean scores from the experimental group while the controlled group's mean score stayed consistent. These results aligned with a study led by Chang (2022), showing that the mean of post test scores of the experimental group were significantly higher than those in the control group. Furthermore, nutritional knowledge scores of the experimental group were significantly higher than those of the control group in the posttest.

## CONCLUSION AND RECOMMENDATIONS

### Conclusion

The following were the conclusions drawn from the results:

1.) Kainpoints is a fun, interactive game that enhances nutrition knowledge through trivia and quizzes. Its scoring system, featuring a streak bonus, rewards consistent learning, while a matching mini-game boosts memory and focus. With an intuitive UI and engaging mechanics, Kainpoints makes learning about nutrition easy and enjoyable.

2.) The Pre-Test mean scores for the experimental and control groups were interpreted as low, and respectively, suggesting a low level of nutritional knowledge. These results imply that respondents in both groups hardly ever showed nutritional comprehension before any intervention.

3). The results showed that the controlled group's nutritional knowledge stayed the same, with a low Post-Test score. On the other hand, the experimental The group showed improvement, reaching a moderate level with this score. This suggests that the gamification intervention helped boost their understanding of nutrition and gained a higher mean score as a result.

4.) Results from the control group indicate that any variation in students' knowledge about nutrition was probable due to random chance and did not reflect any significant improvement. For the experimental group, the results reflect a noticeable improvement in the understanding of nutrition, suggesting that the intervention worked. This indicates that the intervention positively affected the knowledge of the students compared to the control condition

5.) The control group had a low level of nutritional knowledge post-intervention, reflecting little improvement. However, the experimental group reflected a notable rise in nutritional understanding, reflecting the efficacy of the treatment. This suggests that the intervention was effective in enhancing students' knowledge, whereas the control group reflected little or no change.

**Recommendations**

Based on the findings of this research, the researchers recommend the following:

1. For game developers, we recommend incorporating features such as time limits and explanatory feedback on incorrect answers to enhance both the enjoyment and effectiveness of educational games while ensuring they provide constructive learning experiences. Additionally, having the application evaluated by experts can offer deeper insights and establish a stronger foundation for its development, enabling more impactful and user-centric designs.

2. School administrators may implement educational interventions similarly to gamification learning on not just digital but can be physical such as card and board games. Integrating interactive and practical exercises can assist the participants in learning and implementing nutritional principles more effectively. Regular

evaluation must also be done to assess progress and accordingly modify teaching techniques.

3. Teachers may create engaging lessons that encourage student learning about nutrition. Integrative nutrition interventions should be taken into account by school administrators in order to facilitate the comprehensive development of students. With the implementation of gamification, teachers can improve the levels of health awareness among students in a fun and engaging way which will have a positive impact on the students as well as the general public health.

4. With the implementation of gamification in teaching, teachers may undergo training on how to properly handle this specific method in teaching. Balancing fun and practicality will be key to unlocking the pinnacle of the students' understanding and knowledge.

5. Lastly, for future researchers it is advisable to carry out longitudinal research to evaluate the effectiveness of gamification on nutritional knowledge and dietary behavior over time. Such research will establish whether the short-term benefits achieved are maintained or contingently have any further effects on students' health in the long run. Knowing the impacts of gamification would help stakeholders make effective conclusions concerning the integration and application of these modalities within holistic nutritional education approaches.

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# APPENDICES

## INFORMED CONSENT

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**UNIVERSITY OF THE IMMACULATE CONCEPTION**
**Senior High School Unit**
J. P. Laurel Avenue, Bajada Street, Davao City

**INFORMED CONSENT**

Informed Consent Form for **KAINPONTS: ENHANCING NUTRITIONAL KNOWLEDGE THROUGH GAMIFIED LEARNING APPLICATION**

Name of the Researcher(s): **Kyle Christian Clata, Francine Jaian C. Fernandes, Gallera Alex, Cristian Paul Nagar, and Christian Jacob M. Rosas**

Institution: **University of the Immaculate Conception**

**INTRODUCTION**
*You are invited to participate in a research study conducted by Francine Jaian C. Fernandes, Kyle Christian Clata, Gallera Alex, Cristian Paul Nagar, and Christian Jacob M. Rosas at the University of the Immaculate Conception because you fit the inclusion criteria for the informant of our study. Your participation is completely voluntary. Before deciding whether to participate, please read the information provided and ask about anything unclear. When reading, please take the time to review this consent form, and feel free to discuss it with family or friends. If you choose to participate, you will be asked to sign this form and receive a copy for your records.*

**PURPOSE OF THE STUDY**
*This study aims to develop a mobile interactive educational video game for learning and improving nutritional knowledge. Furthermore, it also would like to measure if a significant change exists in a person's knowledge about nutrition before and after Gamification. We are focusing on this particular research problem in order to understand the major issues linked with low nutritional literacy, particularly in children and adolescents, whose access to unhealthy food options has been increasing. To accomplish this, we propose the development of a new educational tool that would be based on video games and aims to capture the interest of users while facilitating the practical application of important nutritional concepts. We wish to test the capability of gamification to transform health education into a more motivating and enjoyable experience and, in turn, enable better dietary decision making among users. The findings of this study will help address the gap in the literature regarding the use of diverse instructional approaches to the discipline of nutrition and how gamified learning may be leveraged for other educational disciplines which might result in wider use in health education and other fields.*

**STUDY PROCEDURES**
*Before participating in this study, the researcher will provide you with an orientation, potentially using a visual aid like a PowerPoint presentation or a lecture, to explain the informed consent form (ICF). After you fully understand the information, we will ask for your consent to participate. If you agree, you will be asked to complete a test questionnaire, which should take approximately 40 minutes. Please know that you have been selected based on specific criteria. The survey instrument has been verified for accuracy and relevance.*

**POTENTIAL RISKS AND DISCOMFORTS**
*During this process, You may encounter certain risks while answering the test questionnaires, although these are relatively low. The primary risks include confusion and nervousness, which can arise if you find the questions challenging. These risks are unlikely to occur frequently, as the questionnaires do not contain personal or sensitive information. To avoid this risk, We ensure that you understand the questionnaires are not graded, thereby alleviating any pressure to perform well. Additionally, we encourage you to answer based on your current knowledge of nutrition, which helps to generate well-diversified results. You are welcome to skip the test that makes you feel emotional or psychological distress, or you can withdraw as a participant if you feel*

*discomfort upon the entire process. We, the researchers, valued your participation and prioritized your well-being throughout the study.*

**POTENTIAL BENEFITS TO PARTICIPANTS AND/OR TO SOCIETY**

*Our gamification study seeks to improve nutritional knowledge among adolescents, offering important benefits to participants and society. Because teenagers' growth necessitates good nutrition, which can be reinforced through teaching tools, this study will assess how interactive applications impact adolescent nutritional knowledge. By combining practical, interactive applications, students can make better and more informed decisions about nutrition and diets, which is why this tool is valuable for health education. With the combination of strong educational ethics and appropriate nutrition-centric lesson plans, students in Catholic institutions have a better chance of achieving holistic development, assisting in developing better nutrition programs. Interactive applications can also help eradicate poor nutrition in early childhood. The results of our research will be disseminated to a wide range of audiences, including healthcare professionals, educators, and the general public, as well as engaging academic peers. Our findings will be published in public peer-reviewed journals on nutrition and health and presented at relevant conferences.*

**PRIVACY AND CONFIDENTIALITY**

*In adherence to the Data Privacy Act 2012, all information gathered from this research's respondents will be kept confidential unless it is necessary to reveal and defend your rights as respondents. The researchers ensure that we will prevent the details of your research involvement from getting out to anyone not connected to the study. No personal identifying information will be presented in any publication or conference presentation discussion of the research findings.*

**PARTICIPATION AND WITHDRAWAL**

*Your participation in this study is voluntary and may be withdrawn at any time without penalty. Your decision not to participate will not involve any negative consequences or loss of benefits to which you would otherwise be entitled. You have the right to withdraw your consent at any point. Participating in this research project will not result in losing legal claims, rights, or remedies.*

**INVESTIGATOR'S CONTACT INFORMATION**

*If you have any questions or concerns about this study, please do not hesitate to contact the researcher of the study through mobile phone number 09936501246 or through email at ffernandes_2023000322@uic.edu.ph if you need to see her; she can be located at Room 301 in the University of the Immaculate Conception, Bonifacio Campus, Davao City.*

**RIGHTS OF RESEARCH RESPONDENTS**

*If you would like to speak with someone outside of the research team or if you have any questions, concerns, or clarifications regarding your rights as research respondents or the research study and are unable to reach the research team, please contact the University of the Immaculate Conception with this number 222-23-20 local 353.*

UNIVERSITY OF THE IMMACULATE CONCEPTION
Senior High School Unit
J. P. Laurel Avenue, Bajada Street, Davao City

**RESEARCH RESPONDENT'S CONSENT**

I have read the information provided above. I have been given a chance to ask questions. My questions have been answered to my satisfaction, and I agree to participate in this study. I have been given a copy of this form. I can withdraw my consent at any time and discontinue participation without penalty.

_________________________________ ________________

**Signature above Printed Name of Participant** **Date Signed**

**To be accomplished by the Researcher Obtaining Consent:**

I have explained the research to the participant and answered all of his/her questions. I believe that he/she understands the information described in this document and freely consents to participate.

_________________________________ ________________

**Name of Person Obtaining Consent** **Date Signed**

# LETTER FOR VALIDATOR

**UNIVERSITY OF THE IMMACULATE CONCEPTION**
**Senior High School**
Bonifacio, Corner De Jesus St, Poblacion District,
Davao City, Davao del Sur 8000

March 10, 2025

**PROF. LILIBETH R. LOZADA**
College of Teacher Education Faculty
University of the Immaculate Conception

Dear Prof. Lozada:

Greetings, Praise be Jesus and Mary!

The undersigned is the group 1 representative of St. James at the University of the Immaculate Conception, undertaking research titled "**KAINPONTS: ENHANCING NUTRITIONAL KNOWLEDGE THROUGH GAMIFIED LEARNING APPLICATION.**"

In connection with this, we request your expertise to validate the attached test questionnaire for the study to ensure that the instrument meets the validity criteria before giving it to the study participants.

Attached here are the research questions, research instrument, and validation sheet. Your feedback on the questions' clarity, relevance, and appropriateness would significantly contribute to refining my research tool.

I highly anticipate your positive response on this matter. Thank you!

Respectfully yours,

**FRANCINE JAIAN C. FERNANDES**
Researcher

Noted by:

**PROF. FRANCIS KENNETH D. CANONO**
Adviser

# VALIDATION FORM

University of the Immaculate Conception
SENIOR HIGH SCHOOL UNIT

## VALIDATION FORM FOR THE QUANTITATIVE QUESTIONNAIRE

Name of Evaluator : Lilibeth Lozada
Degree : MAEM
Position : Faculty
Number of Years in Teaching : 27

**To the Evaluator**: Please check the appropriate box for your rating:

Point Equivalent: 5 - Excellent
4 - Very Good
3 - Good
2 - Fair
1 - Poor

| | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| 1.) **Clarity of Directions and Items.** The vocabulary level, language structure and conceptual of the questions suit the level of the respondents. The test direction and items are written in a clear and understandable manner. | / | | | | |
| 2.) **Presentations / Organization of Items** The items are organized in logical manner. | | / | | | |
| 3.) **Suitability of Items** The items appropriately represent the substance of the research. The questions are designed to determine the conditions, knowledge, skills and attitudes that are supposed to be measured. | / | | | | |
| 4.) **Adequateness of Items per Category** The items represent the coverage of the research adequately. The number of questions per category is representative enough of all questions needed for the research. | | / | | | |
| 5.) **Attainment of the Purpose** The instrument as a whole fulfills the objective for which it was constructed | / | | | | |
| **6.) Objectivity** Each item question only one specific answer or measure only one behavior and no aspect of the questionnaire suggests bias on the part of the research. | / | | | | |
| **7.) Scale and Evaluation in Rating System** The scale adopted is appropriate for the items. | / | | | | |

**Remarks**:
The **quantitative questionnaire meets all validation criteria** and is well-structured to assess the research objective.

Lilibeth Lozada
Signature over Printed Name

# TABLE OF SPECIFICATION
(For Pre and Post Test)

UNIVERSITY OF THE IMMACULATE CONCEPTION
Senior High School

TABLE OF SPECIFICATION
Post-Test

| COMPETENCIES | Remembering | | Understanding | | Applying | | Analyzing | | Evaluating | | Total | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Items | Points | Items | Points | Items | Points | Items | Points | Items | Points | Items | Points |
| **Understanding Vitamins and Minerals** | 4,5,7 | 3 | 9,10,14 | 3 | 15,20,21 | 3 | 27,28 | 2 | 29,33 | 2 | | 13 |
| **Understanding Macronutrient and Micronutrient** | 1,2 | 2 | 11,13 | 2 | 16,18 | 2 | 24,26 | 2 | 30,31,35 | 3 | | 11 |
| **Understanding Saturated and Unsaturated Fats** | 3,6 | 2 | 8,12 | 2 | 17,19 | 2 | 22,23,25 | 3 | 32,34 | 2 | | 11 |
| TOTAL | | 7 | | 7 | | 7 | | 7 | | 7 | | 35 |
| Percentage | | 20% | | 20% | | 20% | | 20% | | 20% | | 100% |

| | |
|---|---|
| Remembering: can the student recall or remember the information? | cite, define, describe, identify, label, list, match, name, outline, quote, recall, report, reproduce, retrieve, show, state, tabulate, and tell. |
| Understanding: can the student explain ideas or concepts? | abstract, arrange, articulate, associate, categorize, clarify, classify, compare, compute, conclude, contrast, defend, diagram, differentiate, discuss, distinguish, estimate, exemplify, explain, extend, extrapolate, generalize, give examples of, illustrate, infer, interpolate, interpret, match, outline, paraphrase, predict, rearrange, reorder, rephrase, represent, restate, summarize, transform, and translate |
| Applying: can the student use the information in a new way? | apply, calculate, carry out, classify, complete, compute, demonstrate, dramatize, employ, examine, execute, experiment, generalize, illustrate, implement, infer, interpret, manipulate, modify, operate, organize, outline, predict, solve, transfer, translate, and use. |
| Analyzing: can the student distinguish between the different parts? | analyze, arrange, break down, categorize, classify, compare, connect, contrast, deconstruct, detect, diagram, differentiate, discriminate, distinguish, divide, explain, identify, integrate, inventory, order, organize, relate, separate, and structure. |
| Evaluating: can the student justify a stand or decision? | appraise, apprise, argue, assess, compare, conclude, consider, contrast, convince, criticize, critique, decide, determine, discriminate, evaluate, grade, judge, justify, measure, rank, rate, recommend, review, score, select, standardize, support, test, and validate. |
| Creating: can the student create a new product or point of view? | assemble, build, collect, combine, compile, compose, constitute, construct, create, design, develop, devise, formulate, generate, hypothesize, integrate, invent, make, manage, modify, organize, perform, plan, prepare, produce, propose, rearrange, reconstruct, reorganize, revise, rewrite, specify, synthesize, and write. |

**TEST QUESTIONNAIRES: PRE-TEST**

**KAINPOINTS: ENHANCING NUTRITIONAL KNOWLEDGE THROUGH GAMIFIED LEARNING APPLICATION**
***PRE-TEST***

Name: ________________________________*(Optional)*
Date: _______
Grade level: ____________________
Age: _______

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**Directions:**
**The following test items follows a multiple-choice test type. For each question, encircle which is the correct answer choice.**

*** Use only Black or Blue ballpen only.**
*** This test is not graded, Answer these questions using your current knowledge.**

*Result of this test will be confidential and will only be used for research purposes.*

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1. Which of the following is a macronutrient?

   A. Vitamin C
   B. Iron.
   C. <u>Protein</u>
   D. Calcium

2. Which nutrient is the body's primary source of energy?

   A. <u>Carbohydrates</u>
   B. Fiber
   C. Fats
   D. Vitamins

3. Which of the following foods is a major source of saturated fat?

   A. Olive oil
   B. <u>Butter</u>
   C. Avocado
   D. Salmon

4. Which vitamin is known as the "sunshine vitamin"?

   A. <u>Vitamin C</u>
   B. Vitamin D
   C. Vitamin A
   D. Vitamin E

5. Which nutrient is essential for the formation and maintenance of healthy bones and teeth?

   A. Iron.
   B. Zinc
   C. <u>Calcium</u>
   D. Potassium

6. Which of the following is a good source of omega-3 fatty acids?

   A. Avocado
   B. Olive Oil
   C. <u>Salmon</u>

D. Almonds

7. Which nutrient is important for healthy vision?

   A. Vitamin D
   B. Vitamin C
   C. Vitamin E
   D. Vitamin A

8. Why are unsaturated fats considered healthier than saturated fats?

   A. They increase bad cholesterol levels in the body.
   B. They provide more energy than saturated fats.
   C. They are easier to digest than saturated fats.
   D. They help reduce bad cholesterol and improve heart health.

9. How does vitamin D contribute to overall health?

   A. It enhances cognitive function
   B. It supports bone health
   C. It boosts immune function
   D. It aids in digestion

10. What role do antioxidants play in the body?

   A. Boosting immune function
   B. Promoting digestion
   C. Preventing cellular damage
   D. Enhancing cognitive function

11. What is the main function of carbohydrates in the body?

   A. Building and repairing body tissues
   B. Energy production
   C. Enhancing immune function
   D. Regulating hormone levels

12. How does replacing saturated fats with unsaturated fats impact heart health?

   A. It increases the risk of heart disease.
   B. It has no effect on cholesterol levels.
   C. It reduces the risk of heart disease by lowering bad cholesterol levels.
   D. It raises both good and bad cholesterol levels equally.

13. What is the main function of proteins in the body?

   A. Providing energy
   B. Building and repairing body tissues
   C. Regulating blood sugar levels
   D. Enhancing cognitive function

14. What is the role of potassium in the body?

   A. Regulating fluid balance
   B. Enhancing muscle function
   C. Promoting bone health
   D. Boosting immune function

15. Emily has a cold and wants to boost her immune system. Which vitamin should she focus on consuming more of?

   A. Foods containing Vitamin A
   B. Foods containing Vitamin C
   C. Foods containing Vitamin D
   D. Foods containing Vitamin E

16. Evere is preparing for a marathon and needs to optimize energy levels. Which nutrient should she prioritize in her diet?
   A. Proteins for muscle repair
   B. Fats for endurance

C. <u>Carbohydrates for sustained energy</u>
D. Fiber for digestive health

17. Charles trying to lose weight is considering different dietary approaches. Which strategy would be most effective for sustainable weight loss?

A. Reducing carbohydrate intake drastically
B. Increasing protein intake to boost metabolism
C. <u>Maintaining a balanced diet with portion control</u>
D. Fasting intermittently

18. Stuart has diabetes and needs to manage his blood sugar levels. Which dietary choice would help him achieve his goal?

A. Simple carbohydrates like sugar
B. Fiber-rich carbohydrates like oats
C. <u>Complex carbohydrates like whole grains</u>
D. Starchy carbohydrates like potatoes

19. Elyza is gluten intolerant and needs to avoid certain foods like protein. Which food group to help manage her condition effectively?

A. Dairy products
B. <u>Wheat and gluten-containing grains</u>
C. Fruits and vegetables
D. Meat and poultry

20. Michael is trying to maintain healthy bones. Which nutrient should he focus on consuming more of?

A. <u>Calcium for bone health</u>
B. Iron for preventing anemia
C. Folic acid for preventing birth defects
D. Vitamin C for immune function

21. Sky is recovering from surgery and needs a diet that supports her wound healing. Which nutrient is essential for this process?

A. <u>Vitamin C for collagen production</u>
B. Vitamin B12 for nerve repair
C. Zinc for immune function
D. Calcium for bone repair

22. How do different types of dietary fats like saturated and unsaturated, and fats. How do these pathways differ in terms of energy production and storage?

A. Proteins are the primary source of energy, carbohydrates for muscle repair, and fats for storage.
B. <u>Carbohydrates are primarily used for immediate energy, proteins for muscle repair, and fats for long-term energy storage.</u>
C. All macronutrients are metabolized similarly for energy production.
D. None of these statements are correct.

23. Which of the following best describes the difference between saturated and unsaturated fats?

A. Saturated fats are liquid at room temperature, while unsaturated fats are solid.
B. <u>Saturated fats are solid at room temperature, while unsaturated fats are liquid.</u>
C. Saturated fats are primarily found in plant-based foods, while unsaturated fats are found in animal products.
D. Saturated fats are more easily metabolized than unsaturated fats.

24. Which of the following best describes the difference between simple and complex carbohydrates fats in terms of blood sugar

levels and overall health?

A. Simple carbohydrates raise blood sugar faster but provide more energy.
B. <u>Complex carbohydrates are digested slowly, reducing the risk of diabetes.</u>
C. Both types have the same effect on blood sugar levels.
D. Simple carbohydrates are better for athletes due to quick energy release.

25. How do different types of dietary fats like saturated and unsaturated fats be more beneficial for heart health and why?

A. Saturated fats are more beneficial due to their energy density.
B. <u>Unsaturated fats are more beneficial as they lower cholesterol levels.</u>
C. Both types of fats have equal health benefits.
D. Saturated fats are better for brain function.

26. Compare the metabolic pathways of carbohydrates, proteins, and fats. How do these pathways differ in terms of energy production and storage?

A. <u>Carbohydrates are primarily used for immediate energy, proteins for muscle repair, and fats for long-term energy storage.</u>
B. Proteins are the primary source of energy, carbohydrates for muscle repair, and fats for storage.
C. All macronutrients are metabolized similarly for energy production.
D. Metabolic pathways do not differ significantly among macronutrients.

27. Compare the roles of vitamins and minerals in maintaining healthy bones. Which nutrients are most crucial and why?

A. <u>Vitamin D and calcium are essential for bone health due to their role in bone mineralization.</u>
B. Vitamin C and iron are more important for bone health due to their antioxidant properties.
C. All vitamins and minerals are equally important for bone health.
D. Vitamins and minerals have no significant impact on bone health.

28. Compare the roles of vitamins A and C in the human body. How do they differ in terms of their primary functions?
A. <u>Vitamin A is primarily for vision, while Vitamin C is for immune function</u>
B. Vitamin A is for immune function, while Vitamin C is for vision
C. Both vitamins are essential for skin health
D. Both vitamins are crucial for energy production

29. Which of the following statements about the health benefits of vitamin C is most accurate?

A. Vitamin C is essential for bone health.
B. <u>Vitamin C supports immune function and collagen production.</u>
C. Vitamin C is primarily important for vision.
D. Vitamin C is crucial for energy production.

30. * "Legumes are a good source of protein and fiber, making them a nutritious addition to a vegetarian diet."* Which of the following best supports this claim?

A. Legumes are high in saturated fats.
B. Legumes are rich in simple carbohydrates.
C. <u>Legumes provide essential amino acids and fiber</u>

D. Legumes are low in essential vitamins.

31. Which of the following statements about the role of carbohydrates in the body is most accurate?

A. Carbohydrates are mainly used for building and repairing body tissues.
B. Carbohydrates are the body's main source of energy.
C. Carbohydrates are essential for immune function.
D. Carbohydrates are crucial for bone health.

32. Which of the following statements about saturated fat is most accurate?

A. Saturated fat is always harmful to health and should be avoided completely.
B. Saturated fat can raise LDL (bad cholesterol), but moderate consumption may not harm overall health.
C. Saturated fat is healthier than unsaturated fat for heart health.
D. Saturated fat has no effect on cholesterol or heart health.

33. Which of the following statements about the health benefits of vitamin D is most accurate?

A. Vitamin D is essential for immune function and energy production.
B. Vitamin D supports bone health and immune function.
C. Vitamin D is primarily important for vision.
D. Vitamin D is crucial for heart health.

34. "Fats are essential for the absorption of certain vitamins." Which type of fat is best described to this claim?

A. Saturated fats for energy production
B. Unsaturated fats for heart health
C. Omega-3 fatty acids for brain function
D. All types of fats aid in vitamin absorption

35. *Evaluate the statement: "Fruits are a significant source of carbohydrates, which are essential for energy production." Which of the following best supports this statement?

A. Fruits contain high amounts of fiber, which aids digestion.
B. Fruits are rich in simple carbohydrates, providing quick energy.
C. Fruits are primarily composed of proteins, which build muscle.
D. Fruits are high in lipids, which support heart health.

**TEST QUESTIONNAIRES: POST-TEST**

**KAINPOINTS: ENHANCING NUTRITIONAL KNOWLEDGE THROUGH GAMIFIED LEARNING APPLICATION**
***POST-TEST***

Name: ________________________________*(Optional)* Date: ________
Grade level: ______________________ Age: ________

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**Directions:**
**The following test items follows a multiple-choice test type. For each question, encircle which is the correct answer choice.**

*** Use only Black or Blue ballpen only.**
*** This test is not graded, Answer these questions using your current knowledge.**

*Result of this test will be confidential and will only be used for research purposes.*

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1. Which nutrient is the body's primary source of energy?

   A. <u>Carbohydrates</u>
   B. Fiber
   C. Fats
   D. Vitamins

2. How do different types of dietary fats like saturated and unsaturated fats be more beneficial for heart health and why?

   A. Saturated fats are more beneficial due to their energy density.
   B. <u>Unsaturated fats are more beneficial as they lower cholesterol levels.</u>
   C. Both types of fats have equal health benefits.
   D. Saturated fats are better for brain function.

3. Which of the following foods is a major source of saturated fat?

   A. Olive oil
   B. <u>Butter</u>
   C. Avocado
   D. Salmon

4. Stuart has diabetes and needs to manage his blood sugar levels. Which dietary choice would help him achieve his goal?

   A. Simple carbohydrates like sugar
   B. Fiber-rich carbohydrates like oats
   C. <u>Complex carbohydrates like whole grains</u>
   D. Starchy carbohydrates like potatoes

5. Which of the following statements about the health benefits of vitamin C is most accurate?

   A. Vitamin C is essential for bone health.
   B. <u>Vitamin C supports immune function and collagen production.</u>
   C. Vitamin C is primarily important for vision.
   D. Vitamin C is crucial for energy production.

6. Which of the following is a good source of omega-3 fatty acids?

   A. Avocado
   B. Olive Oil
   C. Salmon
   D. Almonds

7. Why are unsaturated fats considered healthier than saturated fats?

   A. They increase bad cholesterol levels in the body.
   B. They provide more energy than saturated fats.
   C. They are easier to digest than saturated fats.

9. What is the main function of carbohydrates in the body?

   A. Building and repairing body tissues
   B. Energy production
   C. Enhancing immune function
   D. Regulating hormone levels

10. Which of the following statements about the health benefits of vitamin D is most accurate?

   A. Vitamin D is essential for immune function and energy production.
   B. Vitamin D supports bone health and immune function.
   C. Vitamin D is primarily important for vision.
   D. Vitamin D is crucial for heart health.

11. "Fats are essential for the absorption of certain vitamins." Which type of fat is best described to this claim?

   A. Saturated fats for energy production
   B. Unsaturated fats for heart health
   C. Omega-3 fatty acids for brain function

   D. They help reduce bad cholesterol and improve heart health.

8. "Fats are essential for the absorption of certain vitamins." Which type of fat is best described to this claim?

   A. Saturated fats for energy production
   B. Unsaturated fats for heart health
   C. Omega-3 fatty acids for brain function
   D. All types of fats aid in vitamin absorption

   D. All types of fats aid in vitamin absorption

12. How does replacing saturated fats with unsaturated fats impact heart health?

   A. It increases the risk of heart disease.
   B. It has no effect on cholesterol levels.
   C. It reduces the risk of heart disease by lowering bad cholesterol levels.
   D. It raises both good and bad cholesterol levels equally.

13. What is the main function of proteins in the body?

   A. Providing energy
   B. Building and repairing body tissues
   C. Regulating blood sugar levels
   D. Enhancing cognitive function

14. Emily has a cold and wants to boost her immune system. Which vitamin should she focus on consuming more of?

   A. Foods containing Vitamin A
   B. Foods containing Vitamin C
   C. Foods containing Vitamin D

D. Foods containing Vitamin E

15. How does vitamin D contribute to overall health?

    A. It enhances cognitive function
    B. <u>It supports bone health</u>
    C. It boosts immune function
    D. It aids in digestion

16. Evere is preparing for a marathon and needs to optimize energy levels. Which nutrient should she prioritize in her diet?
    A. Proteins for muscle repair
    B. Fats for endurance
    C. <u>Carbohydrates for sustained energy</u>
    D. Fiber for digestive health

17. Which of the following is a macronutrient?

    A. Vitamin C
    B. Iron.
    C. <u>Protein</u>
    D. Calcium

18. Evaluate the statement: "Fruits are a significant source of carbohydrates, which are essential for energy production." Which of the following best supports this statement?

    A. Fruits contain high amounts of fiber, which aids digestion.
    B. <u>Fruits are rich in simple carbohydrates, providing quick energy.</u>
    C. Fruits are primarily composed of proteins, which build muscle.
    D. Fruits are high in lipids, which support heart health.

19. Which nutrient is important for healthy vision?

    A. Vitamin D
    B. Vitamin C
    C. Vitamin E
    D. <u>Vitamin A</u>

20. How do different types of dietary fats like saturated and unsaturated, and fats. How do these pathways differ in terms of energy production and storage?

    A. Proteins are the primary source of energy, carbohydrates for muscle repair, and fats for storage.
    B. <u>Carbohydrates are primarily used for immediate energy, proteins for muscle repair, and fats for long-term energy storage.</u>
    C. All macronutrients are metabolized similarly for energy production.
    D. None of these statements are correct.

21. Which of the following best describes the difference between saturated and unsaturated fats?

    A. Saturated fats are liquid at room temperature, while unsaturated fats are solid.
    B. <u>Saturated fats are solid at room temperature, while unsaturated fats are liquid.</u>
    C. Saturated fats are primarily found in plant-based foods, while unsaturated fats are found in animal products.
    D. Saturated fats are more easily metabolized than unsaturated fats.

22. What is the role of potassium in the body?

    A. <u>Regulating fluid balance</u>
    B. Enhancing muscle function
    C. Promoting bone health
    D. Boosting immune function

23. Michael is trying to maintain healthy bones. Which nutrient should he focus on consuming more of?

A. Calcium for bone health
B. Iron for preventing anemia
C. Folic acid for preventing birth defects
D. Vitamin C for immune function

24. Which nutrient is essential for the formation and maintenance of healthy bones and teeth?

    A. Iron.
    B. Zinc
    C. Calcium
    D. Potassium

25. Which of the following best describes the difference between simple and complex carbohydrates fats in terms of blood sugar levels and overall health?

    A. Simple carbohydrates raise blood sugar faster but provide more energy.
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    B. Proteins are the primary source of energy, carbohydrates for muscle repair, and fats for storage.
    C. All macronutrients are metabolized similarly for energy production.
    D. Metabolic pathways do not differ significantly among macronutrients.

27. Sky is recovering from surgery and needs a diet that supports her wound healing. Which nutrient is essential for this process?
    A. Vitamin C for collagen production
    B. Vitamin B12 for nerve repair
    C. Zinc for immune function
    D. Calcium for bone repair

28. Compare the roles of vitamins and minerals in maintaining healthy bones. Which nutrients are most crucial and why?

    A. Vitamin D and calcium are essential for bone health due to their role in bone mineralization.
    B. Vitamin C and iron are more important for bone health due to their antioxidant properties.
    C. All vitamins and minerals are equally important for bone health.
    D. Vitamins and minerals have no significant impact on bone health.

29. Charles trying to lose weight is considering different dietary approaches. Which strategy would be most effective for sustainable weight loss?
    A. Reducing carbohydrate intake drastically
    B. Increasing protein intake to boost metabolism
    C. Maintaining a balanced diet with portion control
    D. Fasting intermittently

30. Compare the roles of vitamins A and C in the human body. How do they differ in terms of their primary functions?
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D. Saturated fat has no effect on cholesterol or heart health.

34. Which vitamin is known as the "sunshine vitamin"?

A. <u>Vitamin C</u>
B. Vitamin D
C. Vitamin A
D. Vitamin E

35. Elyza is gluten intolerant and needs to avoid certain foods like protein. Which food group to help manage her condition effectively?

A. Dairy products
B. <u>Wheat and gluten-containing grains</u>
C. Fruits and vegetables
D. Meat and poultry